

Product Requirements

PetAlert

Team Sunrise Inc

Brief problem statement

Losing a pet is a stressful and time-sensitive situation for families. During emergencies, owners often struggle to quickly gather accurate information, contact relevant agencies, and distribute alerts effectively. There is currently no centralized, structured system that stores all pet-related information and generates organized alerts in one place.

PetFinder aims to solve this problem by providing a mobile application that securely stores pet information in the cloud and enables users to quickly generate and share missing pet alerts. The system ensures persistent data storage, accessibility across devices, and structured communication during emergencies.

In addition to emergency alert functionality, the application also supports everyday pet management through reminders, safety information, and contact management features that assist pet owners in caring for their pets effectively.

System Requirements

- **Operating Systems:** Android 10 or later | iOS 15 or later
- **Hardware:** Smartphone with at least 3 GB RAM and 150 MB free storage
- **Network:**
 - Internet required for email and social-media alerts
 - User Authentication (Login / Signup)
 - Future cloud backups and social-media integrations
 - Core functions (viewing, editing, reminders) available offline
- **Third-Party Services:**
 - Firebase Realtime Database (for cloud storage and authentication)
 - Google Maps SDK (for future location tagging enhancements)
 - Flutter Plugins (`image_picker`, `path_provider`, `shared_preferences`, `flutter_local_notifications`)

User Profile

- Primary users are pet owners responsible for household pets (dogs, cats, birds, reptiles, etc.).
- Users are expected to understand English and basic mobile navigation.
- The intended audience includes both tech-savvy and general users; design follows accessibility standards (large fonts, high-contrast mode, screen-reader support).
- Users should feel confident performing essential tasks—adding pets, updating info, sending alerts in three taps or fewer.

List of Features

Provide a numbered list of features (F1, F2, etc.) that concisely, clearly, and accurately describe that which constitutes your project.

F1- Pet profiles that list one pet's information (name, age, breed, a recent picture of the pet, owner's name, owner's contact information, and emergency contact's information). A different profile must be made for each pet.

F2- Alert system that automatically drafts and delivers an email to the local police station with the missing pet(s) information and sends a notification with the same information the police received to other Pet Finder users in the area.

F3- Notification system that regularly prompts the user to update or check that his or her pet's information is up-to-date.

F4- Photo gallery that allows users to upload more than one picture of his or her pet onto the app.

F5- Recommendations section that displays treats or toys a listed pet in the user's pet profiles might enjoy.

F6- Toxic food infographic that displays and informs users of possible dangerous or poisonous foods for a listed pet in the user's pet profiles.

F7 – Cloud-based pet data storage (Firestore integration)

F8 – Cross-device synchronization

F9 – Regular update prompts

F10 – Informational section (Recommendations & Toxic Food Info)

Functional requirements (user stories)

List the Priority as 1 (High Priority - Critical) to 3 (Low Priority – Would be nice if we have time)

No.	User Story Name	Description	Priority
R1	Pet Profile	As a user, I want to be able to make my own profile for my pet.	1
R2	Owner & Emergency Contact Info	As a user, I want to be able to include my contact information in case of emergency.	1
R3	Missing Pet Alert System	As a user, I want to be able to have an alert system/notification for a missing pet.	1
R4	Regular Update Prompts	As a user, I want to be able to post regular updates	2
R5	Photo Gallery	As a user, I want to upload multiple photos of my pet and store those photos in a gallery	2
R6	Section for Recommendations	As a user, I want the app to suggest treats and toys based on pet's species, breed, and age	3
R7	Info Graphics for Possible Toxic Foods	As a user, I want to view a simple infographic of toxic foods, so that I can quickly learn what to avoid.	2

Non-Functional Requirements

Describe any constraints or cross-cutting characteristics of the system in a manner that is clear, specific, and testable. Each requirement should have a unique identifier (e.g. NF1, NF2,...). Only present those which are applicable to your system. Categories include but are not limited to:

Security

- NF1. Implement Firebase Authentication with optional “Stay Logged In” toggle.
- NF2. Encrypt stored pet and contact data on-device using platform-native secure storage.
- NF3 – Images must be securely stored in Firebase Storage with restricted access rules.

Reliability

- NF1. Implement Firebase Authentication with optional “Stay Logged In” toggle.
- NF2. Encrypt stored pet and contact data on-device using platform-native secure storage.
- NF3- Pet data must persist after app restart and across multiple devices.

Usability

- UF1. All major functions accessible within three taps from the home screen.
- UF2. Consistent theme and iconography across all modules.
- UF3. Clear success/error feedback on every major action (save, share, upload).
- UF4. Supports large text, high-contrast mode, and TalkBack/VoiceOver.
- UF5. Designed for stress-free emergency navigation—minimal prompts, focused layout.

- UF6. Cloud data must synchronize correctly between devices logged into the same account.

Cross-Platform Compatibility

- Maintains equivalent functionality on Android and iOS; layout tested on various resolutions.

Accuracy

- NF7. All generated alerts, posts, and emails must display verified pet details and contact information without truncation or formatting errors.

Sponsor Requirements

I have read and approved the material in this document. If there is no external sponsor, the TA or instructor will sign it for accuracy/scope.

Print Name

Signature

Date